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(54) **GENERATING ENGAGEMENT SCORES USING MACHINE LEARNING MODELS FOR USERS INTERACTING WITH VIRTUAL OBJECTS**

(52) **U.S. Cl.**
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(57) **ABSTRACT**

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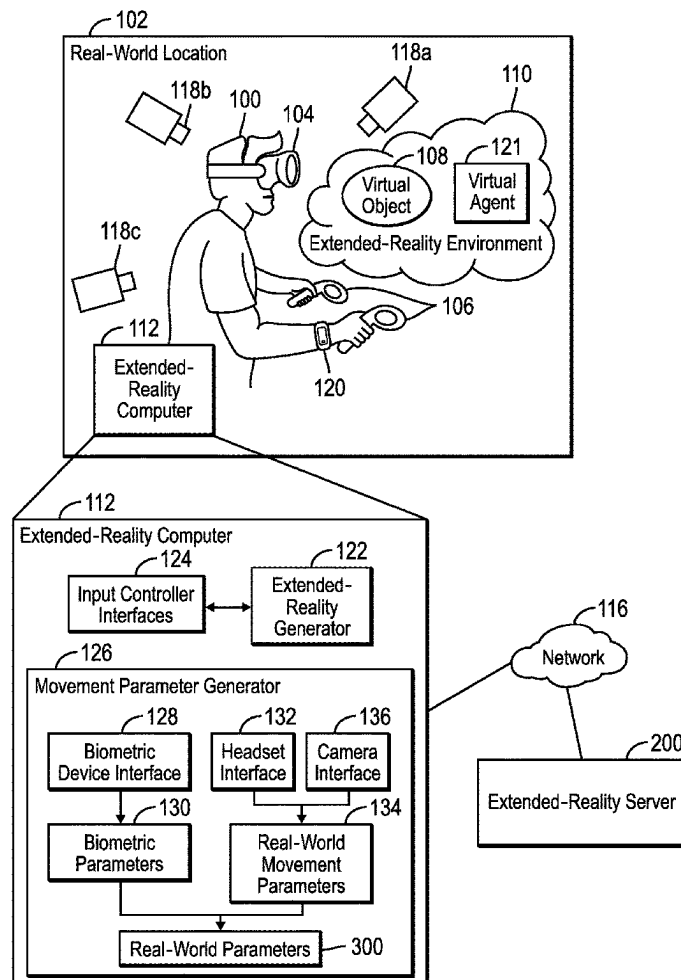
Provided are a computer program product, system, and method for generating engagement scores using machine learning models for users interacting with virtual objects. Movement parameters are received from tracking devices from a user in a real-world while the user is interacting with a virtual object in the extended-reality environment. The movement parameters are processed to determine a first engagement score indicating user interest in the real-world entity represented by the virtual object. A determination is made of on user information access requests with respect to the real-world entity. The information on the user information access requests is inputted to an engagement machine learning model to output a second engagement score indicating user interest in the real-world entity. The first engagement score and the second engagement score are outputted to provide information on an efficacy of the virtual object in promoting interest in the real-world entity.

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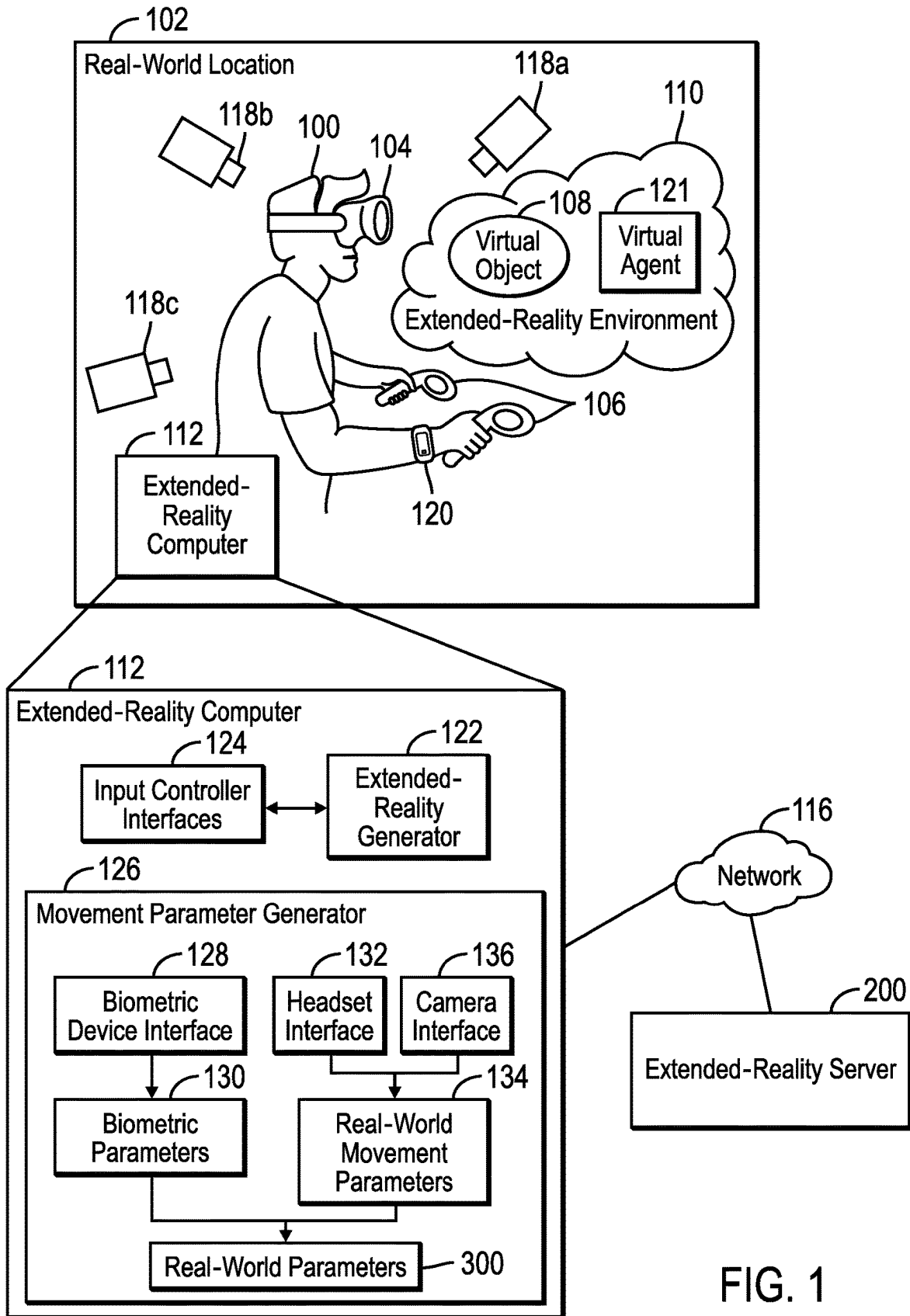


FIG. 1

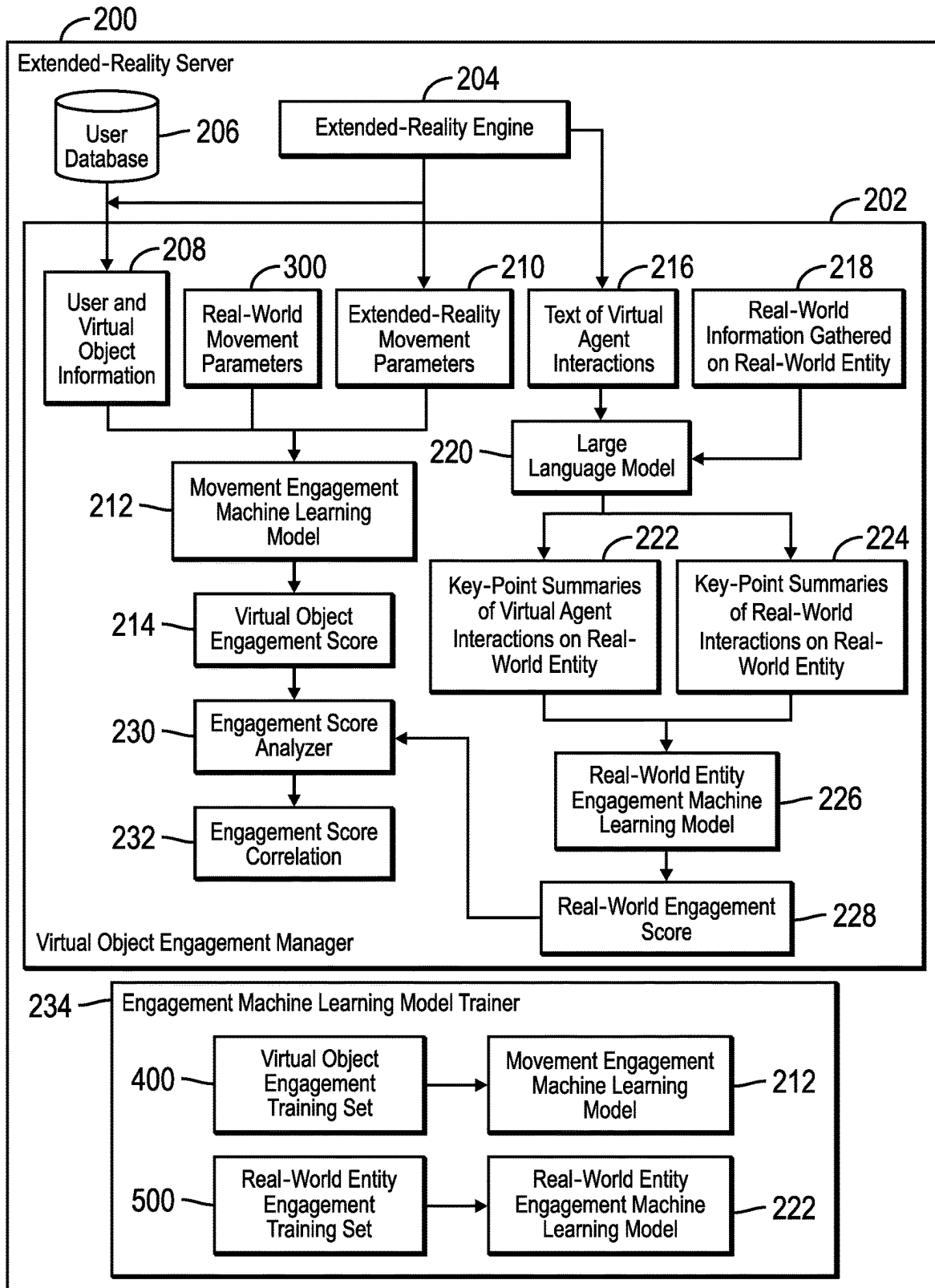


FIG. 2

302	304	306	308	310	312	314	316	318	300
User ID	Engagement Time	Virtual Object ID	Controller Coordinates	Headset Angle	Angular Displacement	Body Speed	Eye Movement	User Biometric Measurements	

Real-World Movement Parameters

FIG. 3

402	404	406	408	410	412	414	416	400 _i
User ID	Virtual Object Info	User Demographics	Real-World Movement Parameters	Extended-Reality Movement Parameters	Predicted Engagement Score	Confidence Level	Real-World Outcome	

Virtual Object Engagement Training Set Instance

FIG. 4

502	504	506	508	510	512	514	516	500 _i
User ID	Virtual Object Info	User Demographics	Key-Point Summaries Of Virtual Agent Interaction	Key-Point Summaries of Real-World Interactions	Predicted Engagement Score	Confidence Level	Real-World Outcome	

Real-World Engagement Training Set Instance

FIG. 5

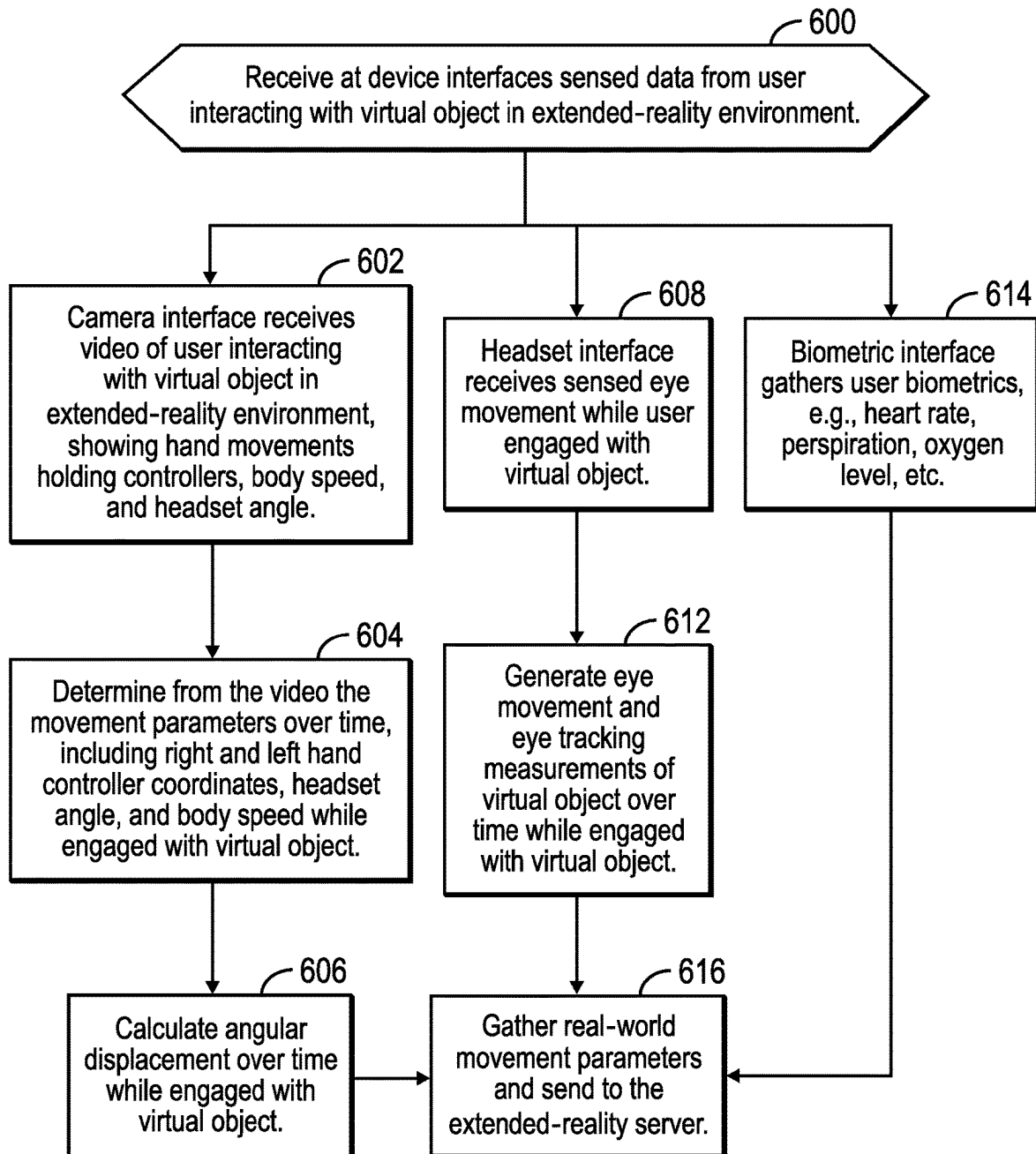


FIG. 6

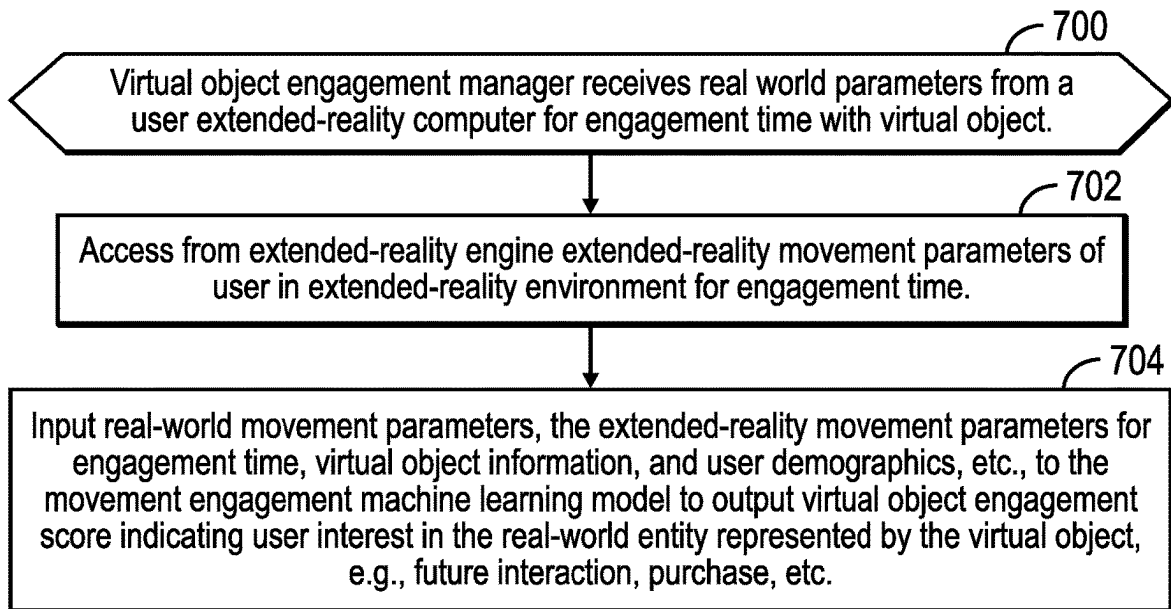


FIG. 7

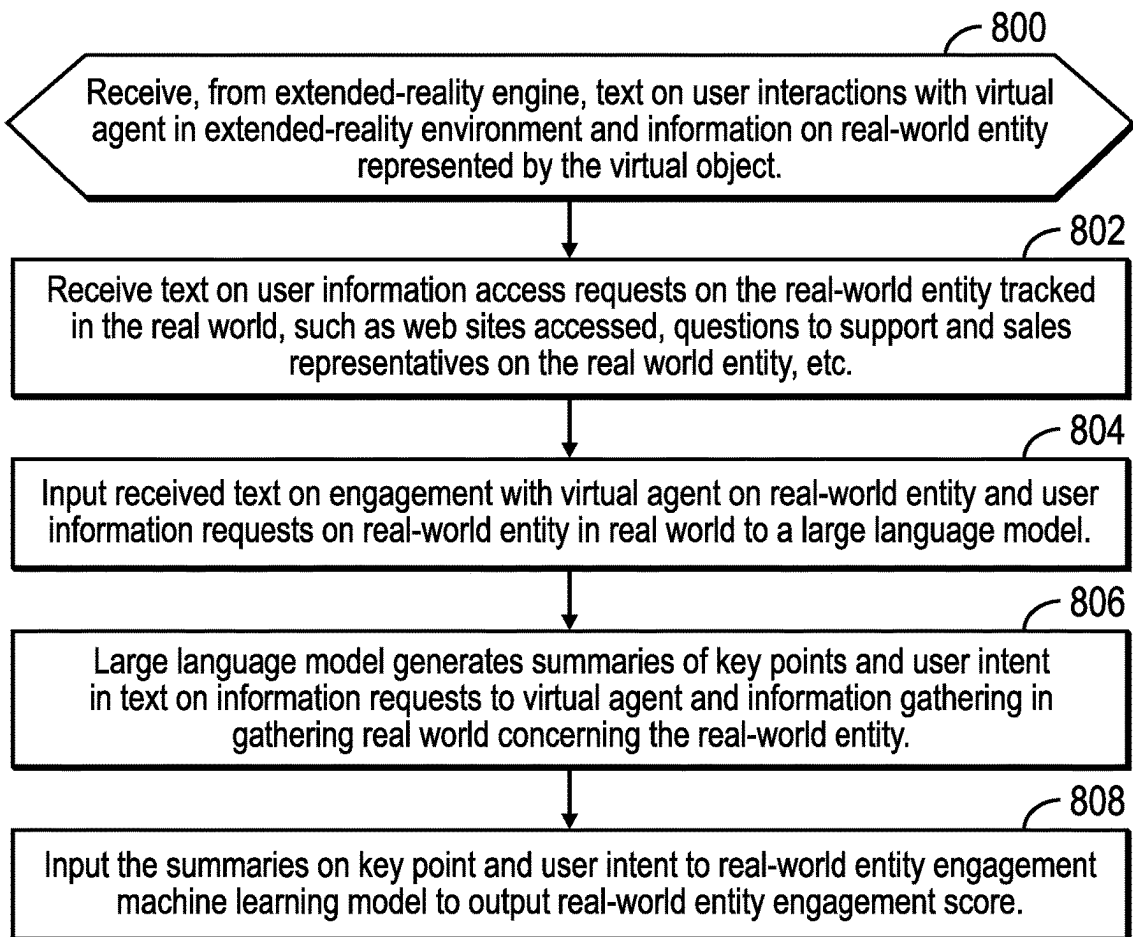


FIG. 8

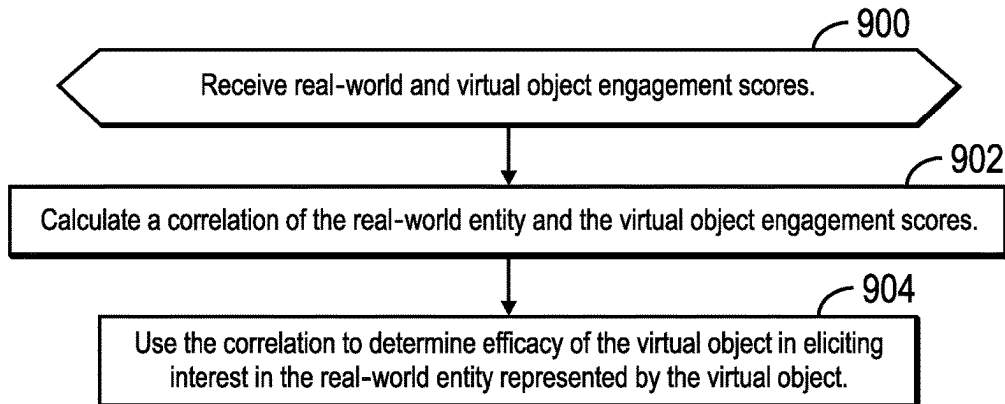


FIG. 9

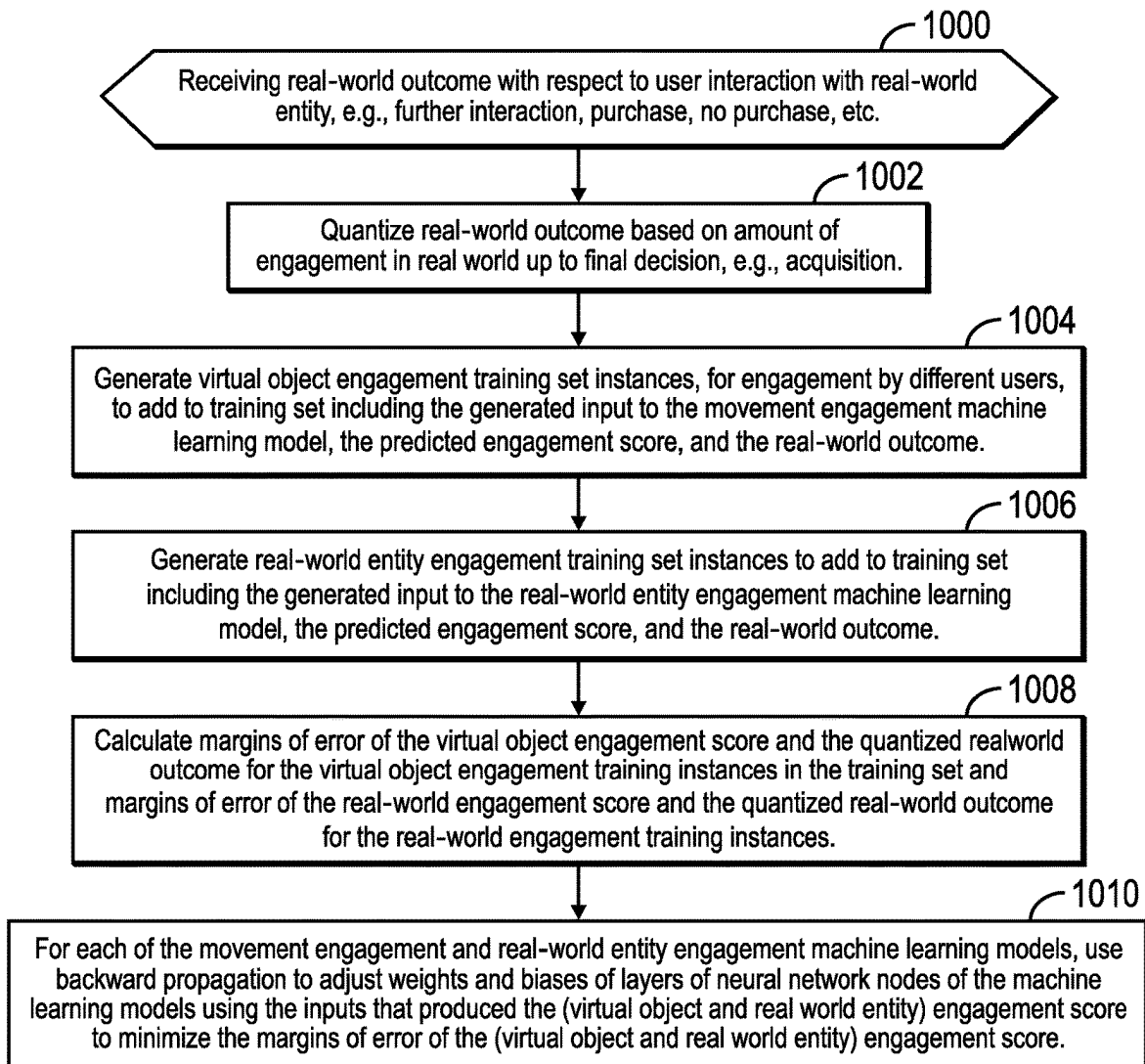


FIG. 10

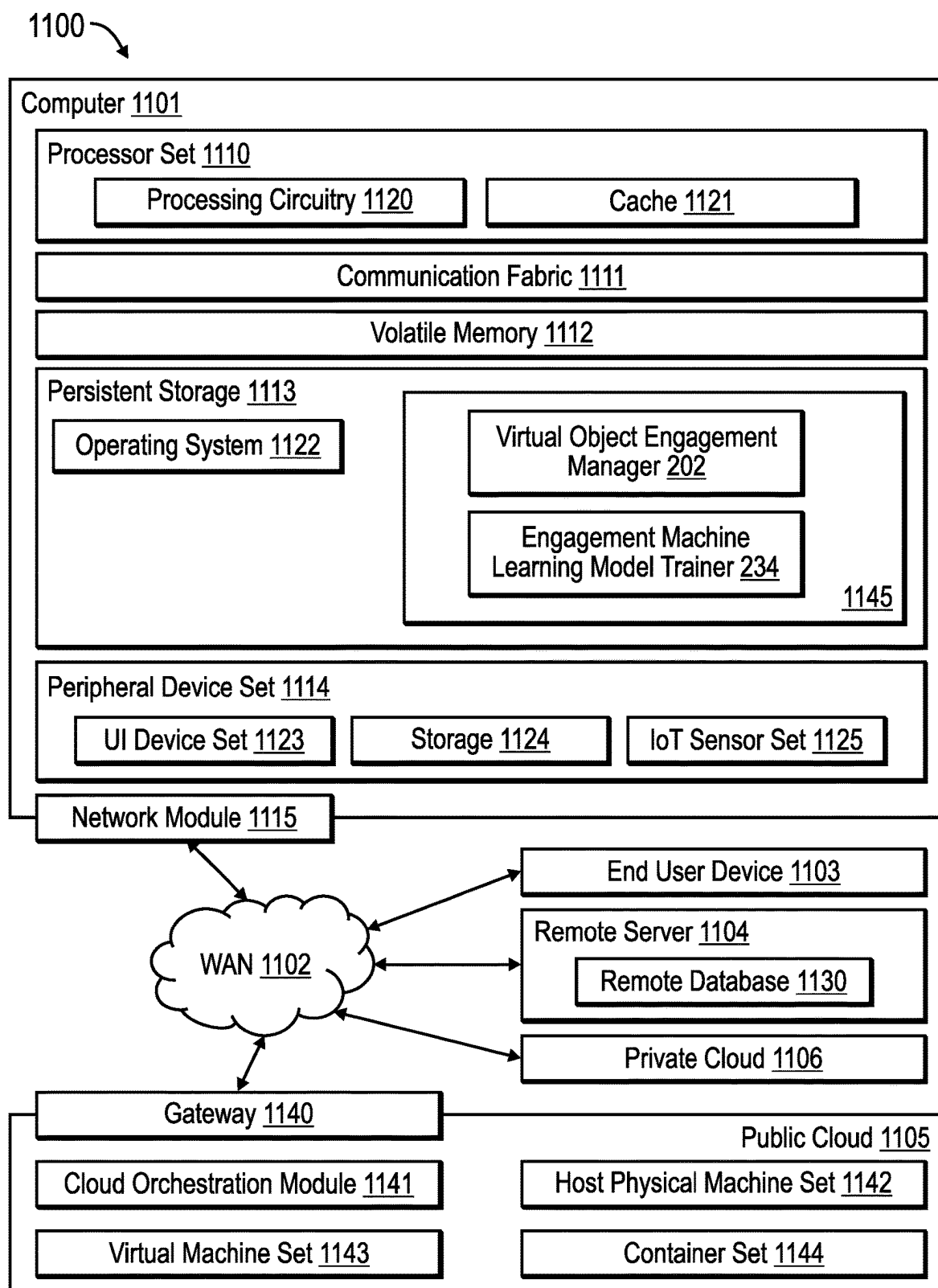


FIG. 11

GENERATING ENGAGEMENT SCORES USING MACHINE LEARNING MODELS FOR USERS INTERACTING WITH VIRTUAL OBJECTS

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to a computer program product, system, and method for generating engagement scores using machine learning models for users interacting with virtual objects.

2. Description of the Related Art

[0002] People may engage in a virtual world, such as a metaverse, using extended-reality headsets to render the virtual world in the user field-of-vision. Participants in the metaverse may also manipulate virtual reality hand controllers to interact with virtual objects in the virtual world. Third parties may have the metaverse provider generate virtual objects into the metaverse representing real-world items of interest to the third parties. Extended-reality (ER) headsets, or glasses, comprise wearable computer-capable glasses or goggles that generate virtual objects, such as three-dimensional images, text, animations, and videos, to overlay into the wearer's field of vision to enable the wearer to view and interact with the virtual objects in the virtual environment.

[0003] Extended reality, as that term is used herein, refers to any of virtual reality (VR), where the entire view of the user is synthetic imagery, augmented reality (AR) where virtual objects or synthetic imagery are added to a view of a real environment, mixed reality (MR) where there is a combination of synthetic and real imagery to form the space, and augmented virtuality (AV) where real imagery is added to a synthetic environment. Thus, extended reality, as that term is used herein, falls on the continuum from total virtuality or total synthetic imagery to a combination of synthetic and real imagery.

SUMMARY

[0004] Provided are a computer program product, system, and method for generating engagement scores using machine learning models for users interacting with virtual objects. Movement parameters are received from tracking devices from a user in a real-world while the user is interacting with a virtual object in the extended-reality environment. The movement parameters are processed to determine a first engagement score indicating user interest in the real-world entity represented by the virtual object. A determination is made of on user information access requests with respect to the real-world entity. The information on the user information access requests is inputted to an engagement machine learning model to output a second engagement score indicating user interest in the real-world entity. The first engagement score and the second engagement score are outputted to provide information on an efficacy of the virtual object in promoting interest in the real-world entity.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 illustrates an embodiment of a computing environment to generate an extended-reality environment in which users may interact with virtual objects.

[0006] FIG. 2 illustrates an embodiment of an extended-reality server to generate engagement scores predicting user interaction with a real-world entity represented by the virtual object with which the user is interacting in the extended-reality environment.

[0007] FIG. 3 illustrates an embodiment of real-world movement parameters detected from the user movement in the real-world while interacting with a virtual object in the virtual world.

[0008] FIG. 4 illustrates an embodiment of a virtual object engagement training set instance to include in a training set to train a movement engagement machine learning model that outputs a predicted user engagement with a real-world entity represented by the virtual object from the movement parameters and other information.

[0009] FIG. 5 illustrates an embodiment of a real-world engagement training set instance to include in a training set to train a real-world engagement machine learning model that outputs a predicted user engagement with a real-world entity based on user information gathering on the real-world entity.

[0010] FIG. 6 illustrates an embodiment of operations to generate real-world movement parameters based on user movement in the real-world while interacting with a virtual object in an extended-reality environment.

[0011] FIG. 7 illustrates an embodiment of operations to use a movement engagement machine learning model to output a virtual object engagement score indicating user likelihood to engage with the real-world entity in the real world.

[0012] FIG. 8 illustrates an embodiment of operations to use a real-world entity engagement machine learning model to output a real-world engagement score indicating user likelihood to engage with the real-world entity in the real world.

[0013] FIG. 9 illustrates an embodiment of operations to analyze the virtual object and real-world engagement scores.

[0014] FIG. 10 illustrates an embodiment of operations to train the movement engagement and the real-world entity engagement machine learning models.

[0015] FIG. 11 illustrates a computing environment in which the components of FIGS. 1 and 2 may be implemented.

DETAILED DESCRIPTION

[0016] Participants in a metaverse may interact with virtual objects in an extended-reality environment that represent real-world entities. Providers of the real-world entities may want to know the efficacy of the virtual objects in promoting user engagement with a real-world entity, represented by the virtual object, in the real-world. Described embodiments provide improvements to computer technology to determine an extent to which user engagement with a virtual object in an extended-reality environment will result in the user engaging with a real-world entity, in the real world, represented by the virtual object. Described embodiments gather movement parameters from the user physical movements and reactions in a real-world location while interacting with a virtual object in the extended-reality world. Described embodiments provide a virtual object machine learning model that may receive as input the gathered movement parameters, based on user corporeal interaction, and output a virtual object engagement score

predicting the extent to which the user will interact with the real-world entity, such as acquire or purchase.

[0017] To validate the virtual object engagement score, described embodiments further gather information on user information access requests for the real-world entity and input to a real-world entity engagement machine learning model to output a second engagement score predicting the extent to which the user will interact with, such as acquire or purchase, the real-world entity. By having two engagement scores outputted by different machine learning models trained on different types of inputs, the engagement scores may validate one another to indicate the validity or usefulness of the engagement score, which reflects on the efficacy of the virtual object in promoting interest in the real-world entity.

[0018] Though this disclosure pertains to the collection of personal data (e.g., real-world movement parameters, extended-reality movement parameters, text of virtual agent interactions, real-world information gathering activities, user information, including demographic information, etc.), it is noted that in embodiments, users opt into the system. In doing so, users (or their guardians) are informed of what data is collected and how it will be used, that any collected personal data may be encrypted while being used, that the users can opt-out at any time, and that if they opt out, any personal data of the user is deleted.

[0019] FIG. 1 illustrates an embodiment of a user 100 in a real-world location 102 wearing an extended-reality headset 104 and manipulating extended-reality hand controllers 106, or other input devices, to interact with a virtual object 108 rendered in a mixed-reality environment 110 in which the user 100 interacts. The virtual object 108 may represent a real-world entity, such as a product, service, item, etc., to provide information on the real-world entity to encourage the user to engage with the real-world entity in the real world. An extended-reality computer 112 renders the mixed-reality environment 110, including objects therein, with which the user may interact. The mixed-reality computer 112 may receive the elements of the mixed-reality environment 110 to render from a mixed reality server 200 over a network 116. The real-world location 102 further includes one or more cameras 118a, 118b, 118c to capture real-world movements of the user 100 while interacting with a virtual object 108 in the extended-reality environment 110. The user 100 may further have wearable sensors 120, such as a biometric gathering device to gather biometric data on the user 100 while engaged with the virtual object 108, such as heart rate, oxygen level, perspiration, steps, movement, etc.

[0020] FIG. 1 shows the components of the extended-reality computer 112 as including an extended-reality generator 122 that renders the extended-reality environment 110 and elements therein, in the headset 104, from extended-reality environment renderings communicated by the extended-reality server 200. The extended-reality generator 122 may generate the virtual object 108 and a virtual agent 121 in the extended-reality environment 110. The virtual agent 121 may comprise a virtual chatbot or other interface rendered in the extended-reality environment 110 to allow the user to request information, such as information on the real-world entity represented by the virtual object 108.

[0021] Input controller interfaces 124, comprising software drivers to receive the commands from the extended-reality controllers 106 and other input devices, provide feedback to the extended-reality generator 122 to control the

user interaction in the extended-reality environment 110. A movement parameter generator 126 includes software drivers to receive input from the devices monitoring the user 100, including a biometric device interface 128 to receive biometric data from the wearable sensors 120 and generate biometric parameters 130 having the gathered biometric information as a function of the engagement time with the virtual object 108, including heart rate, perspiration, oxygen level, etc.; a headset interface 132 to receive information gathered from the headset 104 including gaze tracking information and eye movements to include in the real-world movement parameters 134; and a camera interface 136 to receive video, images and body speed information captured by the cameras 118a, 118b, 118c and generate movement parameters during the engagement time based on the captured images and sensed body speed, including right-hand mixed reality controller handset coordinates, left-hand mixed reality controller handset coordinates, headset angles, and body speeds, during the engagement time, to include in the real-world movement parameters 300.

[0022] The movement parameter generator 126 may further generate derivative information to include in the real-world movement parameters 300, such as angular displacement of the extended-reality hand controllers 106 with respect to the headset 104. For instance, based on a gathered headset angle (θ_t), right-hand controller coordinates (rhc_t), and left-hand controller coordinates (lhc_t) at time t , the angular displacement at time t may be calculated according to equation (1) below:

$$d_t = f(h, rhc, lhc) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} * \left(\frac{\theta}{360} \right) * 2\pi \quad (1)$$

[0023] The movement parameter generator 126 may further calculate another input including a calculated engagement score at time t as a function of the angular displacement (d_t) and a body speed (s_t) at time t , calculated according to equation (2) below, where α comprises an extended-reality constant.

$$d/s * \alpha \quad (2)$$

[0024] The extended reality computer 112 may comprise a computer system embedded in the headset 104, a wearable computer worn by the user 100 or console at the real-world location 102.

[0025] FIG. 2 illustrates an embodiment of the extended-reality server 200 having a virtual object engagement manager 202 to calculate an engagement score with respect to the user interaction with the virtual object indicating a likelihood the user will further interact with or acquire a real-world entity represented by the virtual object 108, such as a product, service or other item. The extended-reality server 200 includes an extended-reality engine 204 to generate mixed-reality environments 110 for users whose information is maintained in a user database 206.

[0026] The virtual object engagement manager 202 gathers user and virtual object information 208 from the user database 206 and extended-reality engine 204, real-world movement parameters 300 provided from the user extended-reality computer 112, and extended-reality movement

parameters **210** having data captured on the user movement in the extended-reality environment **110** from the extended-reality engine **204**. The gathered inputs **208**, **300**, **210** are inputted into a movement engagement machine learning model **212** to generate a virtual object engagement score **214** indicating a likelihood the user will further engage with the real-world entity represented by the virtual object **108**, such as purchase or acquire the real-world entity. The movement engagement machine learning model **212** comprises a machine learning model trained to output a predicted engagement score indicating a likelihood of a further interaction with the real-world entity, such as purchase or acquisition or other engagement action.

[0027] The virtual object engagement manager **202** gathers text of virtual agent interactions **216** with respect to the real-world entity in the extended-reality environment **110** and real-world information **218** the user access on the real-world entity in the real world, such as web sites accessed, inquiries with providers of the real-world entity, etc. The real-world information **218** may be difficult to obtain and require additional monitoring programs, such as use of web browser cookies and add-ins. The text of virtual agent interactions **216** and real-world information **218** are inputted to a large language model **220** to generate key-point summaries and user intent **222** and **224** for the virtual agent interactions **216** and real world gathered information **218**, respectively. The key point summaries and user intent **222** and **224** are then inputted into a real-world entity engagement machine learning model **226** to output a predicted real-world engagement score **228** indicating a likelihood the user will further engage with the real-world entity represented by the virtual object **108**, such as purchase or acquire the real-world entity. The real-world entity engagement machine learning model **226** comprises a machine learning model trained to output a predicted engagement score indicating a likelihood of a further interaction with the real-world entity, such as purchase or acquisition or other engagement action.

[0028] An engagement score analyzer **230** receives the virtual object engagement score **214** and the real-world entity score **228**, and calculates an engagement score correlation **232** between the two engagement scores **228** and **230**. The engagement scores **228** and **230** may be used to confirm the accuracy of each other in predicting the user engagement, such as to purchase the real-world entity product or service. For instance, the real-world engagement score **228**, based on information the user gathers on the real-world entity in the real world, may confirm each other or indicate they cannot be validated. In this way, correlation of the scores may confirm the engagement score.

[0029] The extended-reality server **200** may further include an engagement machine learning model trainer **234** to generate virtual object engagement training set instances **400**; to include in a virtual object engagement training set **400** upon receiving feedback on the real-world outcome for a particular user, such as purchase, not purchase, additional inquiries, etc. The trainer **234** may further generate real-world engagement training set instances **500**; to include in a real-world engagement training set **500** upon receiving feedback on the real-world outcome for a particular user. These training sets **400** and **500** may be used to train the machine learning models **212** and **226**, respectively. In one embodiment, the extended-reality headset **104** may comprise a type of

[0030] computer vision glasses to render extended-reality virtual objects. The extended-reality headset **104** may further comprise a gaze tracking device to receive a gazed virtual object detected by eye tracking cameras that acquire the gazed virtual object on which the tracked eye is fixed and information on coordinates of an axis of a line-of-sight, also referred to as sightline, visual axis, the user is viewing within the field of vision captured by the gaze tracking device tracking. Extended-reality smart glasses are wearable computer-capable glasses that generate virtual objects, such as three-dimensional images, text, animations, and videos, to overlay into the wearer's field of vision so the digital information is viewable along with real-world scenes in the wearer's field of vision. The headset **104** may further provide augmented reality (AR) virtual objects. Augmented reality is used to supplement information presented to users on items they are looking at, such as augmented reality controls to control items in the wearer's field of vision or information on locations in the field of vision. Additionally, the extended-reality headset **104** may provide extended-reality virtual objects that interact with the real-world. For instance, an extended-reality virtual object may react to the user **100** in the same way it would in the real-world, such as appear closer to the user as the user moves closer to the virtual object.

[0031] The extended-reality headset **104** may include a processor, display, sensors and input devices, and may include many of the components found in smartphones and tablet computers. Extended-reality rendering may be performed by optical projection systems, monitors, handheld devices, and display systems worn on the human body. A head-mounted display (HMD) is a display device worn on the forehead, such as a harness or helmet-mounted. HMDs place images of both the physical world and virtual objects over the user's field of view. HMDs may employ sensors for six degrees of freedom monitoring that allow the system to align virtual information to the physical world and adjust accordingly with the user's head movements. The HMDs may also implement gesture controls for full virtual immersion.

[0032] Extended-reality displays may be rendered on devices resembling eyeglasses, and employ cameras to intercept real-world view and re-display its augmented view through the eye pieces and devices in which MR imagery is projected through or reflected off the surfaces of the eyewear lens places. Other implementations of MR displays include a head-up display (HUD), which is a transparent display that presents data without requiring users to look away from their usual viewpoints. Extended-reality may include overlaying the information and registration and tracking between the superimposed perceptions, sensations, information, data, and images and some portion of the real-world. Additional extended-reality implementations include contact lenses and virtual retinal display, where α display is scanned directly into the retina of a viewer's eye. EyeTap augmented reality devices capture rays of light that would otherwise pass through the center of the lens of the wearer's eye, and substitutes synthetic computer-controlled light for each ray of real light. The extended-reality headset **104** may further use motion tracking technologies, including digital cameras and/or other optical sensors, accelerometers, GPS, gyroscopes, solid state compasses, radio-frequency identification (RFID).

[0033] Extended reality, as that term is used herein, refers to any of virtual reality (VR), where the entire view of the user is synthetic imagery, augmented reality (AR) where virtual objects or synthetic imagery are added to a view of a real environment, mixed reality (MR) where there is a combination of synthetic and real imagery to form the space, and augmented virtuality (AV) where real imagery is added to a synthetic environment. Thus, extended reality, as that term is used herein, falls on the continuum from total virtuality or total synthetic imagery to a combination of synthetic and real imagery.

[0034] In a virtual reality environment, the extended-reality display 104 renders the entire environment so the participant is fully immersed in the environment 110 without being able to visualize their external environment. A virtual reality environment may comprise an entirely different environment than that in the real-world location 102. Alternatively, a virtual reality environment may render the actual real-world location environment along with virtual elements. In an augmented reality environment, elements of the extended-reality environment 110, such as the virtual object 108 and virtual agent 121, are rendered within the participant's real-world environment, such as superimposed within the real-world environment.

[0035] The network 116 may comprise a network such as a Storage Area Network (SAN), Local Area Network (LAN), Intranet, the Internet, Wide Area Network (WAN), peer-to-peer network, wireless network, arbitrated loop network, etc.

[0036] The arrows shown in FIGS. 1 and 2 between the components and objects in the extended-reality computer 112 and the extended-reality server 200 represent a data flow between the components.

[0037] Generally, program modules, such as the program components 121, 122, 124, 126, 128, 132, 136, 202, 204, 212, 220, 226, 230, 234 of systems 112 and 200, among others, may comprise routines, programs, objects, components, logic, data structures, and so on that perform particular tasks or implement particular abstract data types. The program components and hardware devices of the systems 112 and 200 in FIGS. 1 and 2 may be implemented in one or more computer systems, where if they are implemented in multiple computer systems, then the computer systems may communicate over a network.

[0038] The program components 121, 122, 124, 126, 128, 132, 136, 202, 204, 212, 220, 226, 230, 234, among others, may be accessed by a processor from memory to execute. Alternatively, some or all of the program components 121, 122, 124, 126, 128, 132, 136, 202, 204, 212, 220, 226, 230, 234, among others, may be implemented in separate hardware devices, such as Application Specific Integrated Circuit (ASIC) hardware devices.

[0039] The functions described as performed by the program components 121, 122, 124, 126, 128, 132, 136, 202, 204, 212, 220, 226, 230, 234, among others, may be implemented as program code in fewer program modules than shown or implemented as program code throughout a greater number of program modules than shown.

[0040] The extended-reality computer 112 may comprise a personal computing device, such as a laptop, desktop computer, tablet, smartphone, wearable computer, etc. The server 200 may comprise one or more server class computing devices, or other suitable computing devices. Alternatively,

the extended-reality computer 112 may be embedded in the extended-reality headset 104.

[0041] In described embodiments, the virtual object engagement manager 202 is maintained in the extended-reality server 200. In alternative embodiments, some are all of the components of the virtual object engagement manager 202 may be maintained in the extended-reality computer 112 to perform these operations locally in the user machine.

[0042] Certain of the program components, such as 212, 220, 226, among others, may use machine learning and deep learning algorithms, such as decision tree learning, association rule learning, neural network, inductive programming logic, support vector machines, Bayesian network, Recurrent Neural Networks (RNN), Feedforward Neural Networks, Convolutional Neural Networks (CNN), Deep Convolutional Neural Networks (DCNNs), Generative Adversarial Network (GAN), etc. For artificial neural network program implementations, the neural network may be trained using backward propagation to adjust weights and biases at nodes in a hidden layer to produce their output based on the received inputs. In backward propagation used to train a neural network machine learning module, biases at nodes in the hidden layer are adjusted accordingly to produce the output having specified confidence levels based on the input parameters. The machine learning models 212, 220, 226 may be trained to produce their output for engagement scores and key-point summaries based on the inputs. Backward propagation may comprise an algorithm for supervised learning of artificial neural networks using gradient descent. Given an artificial neural network and an error function, the method may use gradient descent to find the parameters (coefficients) for the nodes in a neural network or function that minimizes a cost function measuring the difference or error between actual and predicted values for different parameters. The parameters are continually adjusted during gradient descent to minimize the error.

[0043] In an alternative embodiment, the components 212, 220, 226 may be implemented not as a machine learning model but implemented using a rules based system to determine the outputs from the inputs. The components 212, 220, 226 may further be implemented using an unsupervised machine learning module, or machine learning implemented in methods other than neural networks, such as multivariable linear regression models.

[0044] Components implemented as a machine learning model may be implemented in programs in memory or in hardware components, such as a hardware accelerator or an inference engine.

[0045] In described embodiments, the extended-reality engine 204 may implement a metaverse rendered at numerous user extended-reality computers 112. In alternative embodiments, the extended-reality engine 204 may render an entertainment, organizational, business, or personal virtual world.

[0046] FIG. 3 illustrates an embodiment of the real-world movement parameters 300 generated by the movement parameter generator 126 and may include: a user identifier (ID) 302 of the user 100; engagement time 304 comprising a time range during which the user 302 engaged with a virtual object 306; controller coordinates 308 indicating coordinates of the extended-reality controllers 106 during the engagement time 304; headset angle 310 of an angle of the headset 104 during the engagement time 304; a derivative value, such as angular displacement 312 calculated from

movement parameters, during the engagement time **304**, including body speed **314** sensed by the cameras **118a**, **118b**, **118c**, eye movement **316** captured by the headset **104** sensors; and user biometric measurements **318** gathered by the sensor device **120**.

[0047] FIG. 4 illustrates an embodiment of a virtual object engagement training set instance **400**; and may comprise: a user ID **402** of the user for which the information was generated; virtual object information **404** on the virtual object **108** such as information **208**, e.g., category, presentation details, etc.; user demographics **406** of the user **402**. The parameters **408**, **410**, and **412** may comprise information **300**, **210**, **214**. A confidence level **414** of the engagement score **412** indicates a likelihood the engagement score **412** is correct; and a quantized real-world outcome **418** of the user engagement, such as certain types of interactions, acquisition, purchase, no purchase, etc. The real-world outcome would be gathered from external sources in the real-world.

[0048] FIG. 5 illustrates an embodiment of a real-world engagement training set instance **500**. The parameters **502**, **504**, **506**, and **516** may comprise the same information as in parameters **402**, **404**, **406**, and **416**, respectively, in FIG. 4. Parameters **508**, **510**, and **512** may comprise the generated information **222**, **224**, and **228**, respectively. A confidence level **514** of the engagement score **512** indicates a likelihood the engagement score **512** is correct.

[0049] FIG. 6 illustrates an embodiment of operations performed by the movement parameter generator components, including the interfaces **128**, **132**, **136**, e.g., software drives. Upon receiving (at block **600**) at the device interfaces **128**, **132**, **136** sensed data from the user **100** interacting with a virtual object **108** in the extended-reality environment **110**, the operations at blocks **602**, **608**, and **614** may be performed in parallel. The camera interface **136** receives (at block **602**) video from the cameras **118a**, **118b**, **118c** of the user **100** in the real-world location **102**, while interacting with the virtual object **108** in the extended-reality environment **110**, showing hand movements holding the controllers **106**, body speed, and headset **104** angle. The camera interface **136** determines (at block **604**), from the captured video and sensed body speed, movement parameters over time, including right and left hand controller coordinates **308**, headset angle **310**, and body speed **314** while the user is engaged with the virtual object **108**. The movement parameter generator **126** may generate (at block **606**) derivative information from the measured parameters, such as angular displacement **312**, physical engagement score, e.g., angular displacement **312** divided by body speed over time.

[0050] The headset interface **132** receives (at block **608**), through sensors in the headset **104**, sensed user **100** eye movement while the user **100** is engaged with the virtual object **108**. The headset interface **132** generates (at block **612**) eye movement and eye tracking measurements of the virtual object **108** over time while the user **100** is engaged with virtual object **108**. The biometric device interface **128** gathers (at block **614**) user biometrics, e.g., heart rate, perspiration, oxygen level, etc. All the gathered information, at blocks **604**, **606**, **612**, and **614**, is stored (at block **616**) in the real-world movement parameters **300** for this virtual object **108** engagement session, and sent to the extended-reality server **200** for processing.

[0051] With the embodiment of FIG. 6, the user extended-reality computer **114** may gather the captured information

related to user **100** movement in the real-world location **102** while interacting with a virtual object **108** in the extended-reality environment **110** for the engagement time.

[0052] FIG. 7 illustrates an embodiment of operations performed by components of the virtual object engagement manager **202** to generate the virtual object engagement score **214**. Upon receiving (at block **700**) the real-world movement parameters **300** from a user extended-reality computer **112** for an engagement time with the virtual object **108**, the virtual object engagement manager **202** accesses (at block **702**), from the extended-reality engine **204**, extended-reality movement parameters **210** of the user **100** virtual movements in the extended-reality environment **110** for the engagement time. The virtual object engagement manager **202** inputs (at block **704**) the real-world movement parameters **300**, the extended-reality movement parameters **210** for the engagement time, virtual object information and user demographics **208**, and other information having high predictive qualities, to the movement engagement machine learning model **212** to output a virtual object engagement score **214** indicating the user **100** interest in the real-world entity represented by the virtual object **108**, e.g., such as predicted future interaction, acquisition, purchase, no purchase, etc.

[0053] FIG. 8 illustrates an embodiment of operations performed by components of the virtual object engagement manager **202** to generate the real-world engagement score **228**. The virtual object engagement manager **202** receives (at block **800**), from extended-reality engine **204**, text on user interactions with virtual agent **216** in the extended-reality environment **110** and information on the on real-world entity represented by the virtual object **108**. The virtual object engagement manager **202** further receives (at block **802**) text on user information access requests on the real-world entity tracked in the real world **218**, such as web sites accessed, questions to support and sales representatives on the real-world entity, etc. The received text on engagement with virtual agent **216** and the user information requests on real-world entity in real world **218** are inputted (at block **804**) to the large language model **220**. The large language model **220** generates (at block **806**) summaries of key points and user intent in text on information requests to virtual agent **121** and information gathering in real-world concerning the real-world entity **224**. These text summaries and information on intent **222** and **224** are inputted (at block **808**) to the real-world entity engagement machine learning model **226** to output a real-world entity engagement score **228**.

[0054] With the described embodiments, manifestations of physical movements of the user **100** at the real-world location **102** are captured and used to generate an engagement score based on the user movement, biometrics, eye movements, etc. Further, at the same time, text of user information access requests to a virtual agent **121** and/or information gathering in the real-world are inputted to a separate machine learning model **226** to generate an engagement score **228** based on user activity with respect to gathering and requesting information on the real-world entity.

[0055] FIG. 9 illustrates an embodiment of operations performed by the engagement score analyzer **230** to further use the two different engagement scores **214** and **228** to glean information on the efficacy of the virtual object **108** in promoting user interest in the real-world entity represented

by the virtual object **108**, including acquisition or purchase of the real-world entity in embodiments when the real-world entity is a product or service and the virtual object **108** comprises an advertisement. Upon the engagement score analyzer **230** receiving (at block **900**) the real-world **228** and virtual object **214** engagement scores, the engagement score analyzer **230** calculates (at block **902**) a correlation of the real-world entity **228** and the virtual object **214** engagement scores indicating whether they correlate or indicate the same level of engagement. The correlation may be used to determine (at block **904**) the efficacy of the virtual object **108** in eliciting interest in the real-world entity represented by the virtual object **108** using a rules based heuristic algorithm.

[0056] For instance, the real-world engagement score **228**, based on information the user gathers on the real-world entity in the real world, may confirm each other or indicate they cannot be validated. For instance, the engagement score analyzer **230** may implement heuristic rules such that if the virtual object engagement score **214** is low, but the real-world engagement score **228** is high, then even though the virtual object **108** failed to affect engagement with the user, the user interest in the real-world entity was nonetheless high. Such a disparity may indicate the virtual object **214** is inadequate and needs to be improved in how it appears to the user in the extended-reality environment **110** to improve engagement in the extended-reality world. Yet further, a high virtual object engagement score **214** but a low real-world engagement score **228** indicates that although the virtual object **108** is appealing and draws in users, it may not be attracting a group of users who would be interested in the real-world entity. In such case, the heuristic rules may suggest to the provider of the virtual object **108** to target the virtual object **108** to different audiences who are more likely to be prospective purchasers of the real-world entity, as opposed to those having no interest to acquire or further engage.

[0057] FIG. **10** illustrates an embodiment of operations performed by the engagement machine learning model trainer **234** to train the machine learning models **212**, **226** to minimize an error between the predicted engagement scores **214**, **228** and the real-world outcome. Upon receiving (at block **1000**) a real-world outcome with respect to user interaction with a real-world entity, e.g., further interaction, purchase, no purchase, etc., the received real-world outcome may be quantized (at block **1002**) to normalize with the engagement score scale based on amount of engagement in real world up to final decision, e.g., acquisition or no acquisition. Virtual object engagement training set instances **400**; are generated (at block **1004**), for engagement by different users, to add to a training set **400** including the generated input **208**, **300**, **210** to the movement engagement machine learning model **212**, the predicted engagement score **214**, and the quantized real-world outcome **416**. Real-world entity engagement training set instances **500**; are generated (at block **1006**) to add to training set **500**, including the generated input **222**, **224** to the real-world entity engagement machine learning model **226**, the predicted engagement score **228**, and the real-world outcome.

[0058] The trainer **234** calculates (at block **1008**) margins of error of the virtual object engagement score **214** and the quantized real-world outcome **416** for the virtual object engagement training set instances **400**; in the training set **400** and margins of error of the real-world engagement score

228 and the quantized real-world outcome **516** for the real-world engagement training set instances **500**; in the training set **500**. For each of the movement engagement **212** and real-world entity engagement **228** machine learning models, the trainer **234** may use (at block **1010**) backward propagation to adjust weights and biases of layers of neural network nodes of the machine learning models **212**, **226**, using the inputs that produced the (virtual object and real world entity) engagement score to minimize the margins of error of the (virtual object and real world entity) engagement score.

[0059] With the embodiment of FIG. **10**, the movement engagement machine learning model **212** is trained to predict an engagement score **214** of a real-world outcome, such as purchase, based on user interactions observed in the real-world while interacting with a virtual object **108** in the extended-reality environment **110**. In this way, one can predict a user ultimate decision, such as purchase, no purchase, with respect to a real-world entity based on their observed corporeal reactions while interacting in the extended-reality environment **110**. Further, user information gathering activities with respect to the real-world entity in the virtual world **110** and real world may separately be used to predict engagement and be trained to predict engagement based on the real-world outcomes.

[0060] The present invention may be a system, a method, and/or a computer program product. The computer program product may include a computer-readable storage medium (or media) having computer-readable program instructions thereon for causing a processor to carry out aspects of the present invention.

[0061] Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

[0062] A computer program product embodiment (“CPP embodiment” or “CPP”) is a term used in the present disclosure to describe any set of one, or more, storage media (also called “mediums”) collectively included in a set of one, or more, storage devices that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A “storage device” is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer-readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded

device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer-readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

[0063] With respect to FIG. 11, computing environment 1100 contains an example of an environment for the execution of at least some of the computer code involved in performing the inventive methods, such as the components in block 1145, including the virtual object engagement manager 202, to determine and analyze a virtual object engagement score 214 and real-world engagement score 228 indicating a user likelihood to further engage with a real-world entity represented by a virtual object 108, and the engagement machine learning model trainer 234 to train the machine learning models 212, 226 used to determine the engagement scores 214, 228. In addition to block 1145, computing environment 1100 includes, for example, computer 1101, wide area network (WAN) 1102, end user device (EUD) 1103, remote server 1104, public cloud 1105, and private cloud 1106. In this embodiment, computer 1101 includes processor set 1110 (including processing circuitry 1120 and cache 1121), communication fabric 1111, volatile memory 1112, persistent storage 1113 (including operating system 1122 and block 1145, as identified above), peripheral device set 1114 (including user interface (UI) device set 1123, storage 1124, and Internet of Things (IoT) sensor set 1125), and network module 1115. Remote server 1104 includes remote database 1130. Public cloud 1105 includes gateway 1140, cloud orchestration module 1141, host physical machine set 1142, virtual machine set 1143, and container set 1144.

[0064] COMPUTER 1101 may take the form of a desktop computer, laptop computer, tablet computer, smart phone, smart watch or other wearable computer, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network or querying a database, such as remote database 1130. As is well understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. On the other hand, in this presentation of computing environment 1100, detailed discussion is focused on a single computer, specifically computer 1101, to keep the presentation as simple as possible. Computer 1101 may be located in a cloud, even though it is not shown in a cloud in FIG. 11. On the other hand, computer 1101 is not required to be in a cloud except to any extent as may be affirmatively indicated.

[0065] PROCESSOR SET 1110 includes one, or more, computer processors of any type now known or to be developed in the future. Processing circuitry 1120 may be distributed over multiple packages, for example, multiple,

coordinated integrated circuit chips. Processing circuitry 1120 may implement multiple processor threads and/or multiple processor cores. Cache 1121 is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set 1110. Cache memories are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located “off chip.” In some computing environments, processor set 1110 may be designed for working with qubits and performing quantum computing.

[0066] Computer-readable program instructions are typically loaded onto computer 1101 to cause a series of operational steps to be performed by processor set 1110 of computer 1101 and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods included in this document (collectively referred to as “the inventive methods”). These computer-readable program instructions are stored in various types of computer-readable storage media, such as cache 1121 and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set 1110 to control and direct performance of the inventive methods. In computing environment 1100, at least some of the instructions for performing the inventive methods may be stored in block 1145 in persistent storage 1113.

[0067] COMMUNICATION FABRIC 1111 is the signal conduction path that allows the various components of computer 1101 to communicate with each other. Typically, this fabric is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up buses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used, such as fiber optic communication paths and/or wireless communication paths.

[0068] VOLATILE MEMORY 1112 is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM. Typically, volatile memory 1112 is characterized by random access, but this is not required unless affirmatively indicated. In computer 1101, the volatile memory 1112 is located in a single package and is internal to computer 1101, but, alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computer 1101.

[0069] PERSISTENT STORAGE 1113 is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of this storage means that the stored data is maintained regardless of whether power is being supplied to computer 1101 and/or directly to persistent storage 1113. Persistent storage 1113 may be a read only memory (ROM), but typically at least a portion of the persistent storage allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage include magnetic disks and solid state storage devices. Operating system 1122 may take several forms, such as various known proprietary operating systems or open source Portable Operating System Interface-type operating systems that employ a kernel. The code included in block 1145 typically includes at least some of the computer code involved in performing the inventive methods.

[0070] PERIPHERAL DEVICE SET **1114** includes the set of peripheral devices of computer **1101**. Data communication connections between the peripheral devices and the other components of computer **1101** may be implemented in various ways, such as

[0071] Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion-type connections (for example, secure digital (SD) card), connections made through local area communication networks and even connections made through wide area networks such as the internet. In various embodiments, UI device set **1123** may include components such as a display screen, speaker, microphone, wearable devices (such as goggles and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **1124** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **1124** may be persistent and/or volatile. In some embodiments, storage **1124** may take the form of a quantum computing storage device for storing data in the form of qubits. In embodiments where computer **1101** is required to have a large amount of storage (for example, where computer **1101** locally stores and manages a large database) then this storage may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. IoT sensor set **1125** is made up of sensors that can be used in Internet of Things applications. For example, one sensor may be a thermometer and another sensor may be a motion detector.

[0072] NETWORK MODULE **1115** is the collection of computer software, hardware, and firmware that allows computer **1101** to communicate with other computers through WAN **1102**. Network module **1115** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In some embodiments, network control functions and network forwarding functions of network module **1115** are performed on the same physical hardware device. In other embodiments (for example, embodiments that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **1115** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer-readable program instructions for performing the inventive methods can typically be downloaded to computer **1101** from an external computer or external storage device through a network adapter card or network interface included in network module **1115**.

[0073] WAN **1102** is any wide area network (for example, the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now known or to be developed in the future. In some embodiments, the WAN **1102** may be replaced and/or supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

[0074] END USER DEVICE (EUD) **1103** is any computer system that is used and controlled by an end user (for example, a customer of an enterprise that operates computer **1101**), and may take any of the forms discussed above in connection with computer **1101**. EUD **1103** typically receives helpful and useful data from the operations of computer **1101**. For example, in a hypothetical case where computer **1101** is designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **1115** of computer **1101** through WAN **1102** to EUD **1103**. In this way, EUD **1103** can display, or otherwise present, the recommendation to an end user. In some embodiments, EUD **1103** may be a client device, such as thin client, heavy client, mainframe computer, desktop computer and so on. The EUD **1103** may comprise the extended reality computer **112** used to render the extended-reality environment **110** for users.

[0075] REMOTE SERVER **1104** is any computer system that serves at least some data and/or functionality to computer **1101**. Remote server **1104** may be controlled and used by the same entity that operates computer **1101**. Remote server **1104** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **1101**. For example, in a hypothetical case where computer **1101** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **1101** from remote database **1130** of remote server **1104**.

[0076] PUBLIC CLOUD **1105** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economies of scale. The direct and active management of the computing resources of public cloud **1105** is performed by the computer hardware and/or software of cloud orchestration module **1141**. The computing resources provided by public cloud **1105** are typically implemented by virtual computing environments that run on various computers making up the computers of host physical machine set **1142**, which is the universe of physical computers in and/or available to public cloud **1105**. The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **1143** and/or containers from container set **1144**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **1141** manages the transfer and storage of images, deploys new instantiations of VCEs and manages active instantiations of VCE deployments. Gateway **1140** is the collection of computer software, hardware, and firmware that allows public cloud **1105** to communicate through WAN **1102**.

[0077] Some further explanation of virtualized computing environments (VCEs) will now be provided. VCEs can be stored as “images.” A new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically

behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can utilize all resources of that computer, such as connected devices, files and folders, network shares, CPU power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

[0078] PRIVATE CLOUD **1106** is similar to public cloud **1105**, except that the computing resources are only available for use by a single enterprise. While private cloud **1106** is depicted as being in communication with WAN **1102**, in other embodiments a private cloud may be disconnected from the internet entirely and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (for example, private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this embodiment, public cloud **1105** and private cloud **1106** are both part of a larger hybrid cloud. CLOUD COMPUTING SERVICES AND/OR MICROSERVICES (not separately shown in FIG. 11): private and public clouds **1106** are programmed and configured to deliver cloud computing services and/or microservices (unless otherwise indicated, the word “microservices” shall be interpreted as inclusive of larger “services” regardless of size). Cloud services are infrastructure, platforms, or software that are typically hosted by third-party providers and made available to users through the internet.

[0079] Cloud services facilitate the flow of user data from front-end clients (for example, user-side servers, tablets, desktops, laptops), through the internet, to the provider’s systems, and back. In some embodiments, cloud services may be configured and orchestrated according to as “as a service” technology paradigm where something is being presented to an internal or external customer in the form of a cloud computing service. As-a-Service offerings typically provide endpoints with which various customers interface. These endpoints are typically based on a set of APIs. One category of as-a-service offering is Platform as a Service (PaaS), where a service provider provisions, instantiates, runs, and manages a modular bundle of code that customers can use to instantiate a computing platform and one or more applications, without the complexity of building and maintaining the infrastructure typically associated with these things. Another category is Software as a Service (SaaS) where software is centrally hosted and allocated on a subscription basis. SaaS is also known as on-demand software, web-based software, or web-hosted software. Four technological sub-fields involved in cloud services are: deployment, integration, on demand, and virtual private networks.

[0080] The letter designators, such as i, among others, are used to designate an instance of an element, i.e., a given element, or a variable number of instances of that element when used with the same or different elements.

[0081] The terms “an embodiment”, “embodiment”, “embodiments”, “the embodiment”, “the embodiments”, “one or more embodiments”, “some embodiments”, and

“one embodiment” mean “one or more (but not all) embodiments of the present invention(s)” unless expressly specified otherwise.

[0082] The terms “including”, “comprising”, “having” and variations thereof mean “including but not limited to”, unless expressly specified otherwise.

[0083] The enumerated listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise.

[0084] The terms “a”, “an” and “the” mean “one or more”, unless expressly specified otherwise.

[0085] Devices that are in communication with each other need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices that are in communication with each other may communicate directly or indirectly through one or more intermediaries.

[0086] A description of an embodiment with several components in communication with each other does not imply that all such components are required. On the contrary a variety of optional components are described to illustrate the wide variety of possible embodiments of the present invention.

[0087] When a single device or article is described herein, it will be readily apparent that more than one device/article (whether or not they cooperate) may be used in place of a single device/article. Similarly, where more than one device or article is described herein (whether or not they cooperate), it will be readily apparent that a single device/article may be used in place of the more than one device or article or a different number of devices/articles may be used instead of the shown number of devices or programs. The functionality and/or the features of a device may be alternatively embodied by one or more other devices which are not explicitly described as having such functionality/features. Thus, other embodiments of the present invention need not include the device itself.

[0088] The foregoing description of various embodiments of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the invention be limited not by this detailed description, but rather by the claims appended hereto. The above specification, examples and data provide a complete description of the manufacture and use of the composition of the invention. Since many embodiments of the invention can be made without departing from the spirit and scope of the invention, the invention resides in the claims herein after appended.

What is claimed is:

1. A computer program product for determining an engagement score of a user for a real-world entity represented by a virtual object in an extended-reality environment, the computer program product comprising a computer readable storage medium having computer readable program code embodied therein that is executable to perform operations, the operations comprising:

receiving, from tracking devices, movement parameters from a user in a real-world while the user is interacting with a virtual object in the extended-reality environment;

processing the movement parameters to determine a first engagement score indicating user interest in the real-world entity represented by the virtual object;
 determining information on user information access requests with respect to the real-world entity;
 inputting the information on the user information access requests to an engagement machine learning model to output a second engagement score indicating user interest in the real-world entity; and
 outputting the first engagement score and the second engagement score to provide information on an efficacy of the virtual object in promoting interest in the real-world entity.

2. The computer program product of claim 1, wherein the operations further comprise:

determining a correlation of the first engagement score and the second engagement score, wherein the correlation is used to determine the efficacy of the virtual object in promoting interest in the real-world entity.

3. The computer program product of claim 1, wherein the tracking devices include a camera to record the movement parameters of the user moving in the real-world and an extended-reality headset to capture user eye movements during interaction with the virtual object in the extended-reality environment, wherein the determining the first engagement score comprises:

inputting the movement parameters and the user eye movement to a first engagement machine learning model to output the first engagement score, wherein the engagement machine learning model to output the second engagement score comprises a second engagement machine learning model.

4. The computer program product of claim 3, wherein the movement parameters include, over an engagement time of the user interaction with the virtual object, right-hand extended-reality controller handset coordinates, left-hand extended-reality controller handset coordinates, headset angles, and body speeds.

5. The computer program product of claim 3, wherein the determining the first engagement score comprises:

calculating angular displacements, during engagement time, as a function of the movement parameters comprising handset controller movement coordinates and headset angles, wherein the movement parameters inputted to the first engagement machine learning model comprise the angular displacements.

6. The computer program product of claim 1, wherein the determining the information on the user information access requests with respect to the real-world entity comprises:

determining user questions entered into a virtual agent rendered in the extended-reality environment concerning the real-world entity; and

processing, by a large language model, the user questions to output summaries of key points of the user questions and user intent with respect to the real-world entity, wherein the summaries of the key points and the user intent are entered into the engagement machine learning model to determine the second engagement score.

7. The computer program product of claim 1, wherein the determining the information on the user information access requests with respect to the real-world entity comprises:

tracking information on the real-world entity the user accessed at computer network locations external to the extended-reality environment, wherein information on

the tracked information is entered into the engagement machine learning model to determine the second engagement score.

8. The computer program product of claim 1, wherein the virtual object comprises an advertisement of the real-world entity comprising an advertised product or service, and wherein the engagement score indicates a likelihood the user will purchase the advertised product or service in the real-world.

9. The computer program product of claim 1, wherein the processing the movement parameters comprises inputting the movement parameters to a first engagement machine learning model to determine the first engagement score, wherein the engagement machine learning model to calculate the second engagement score comprises a second engagement machine learning model, wherein the operations further comprise:

receiving real-world outcomes with respect to users, for which the first and second engagement scores are generated, interacting with the real-world entity in the real world;

generating a first training set including movement parameters for multiple users interacting with the virtual object, first engagement scores for the multiple users, a quantized real-world outcomes, and margins of error between the first engagement scores and the quantized real-world outcomes; and

training the first engagement machine learning model to output first engagement scores that minimize the margins of error.

10. The computer program product of claim 9, wherein the operations further comprise:

generating a second training set including the information on the user information access requests for multiple users interacting with the virtual object, the second engagement scores for the multiple users, the quantized real-world outcomes, and margins of error between the second engagement scores and the quantized real-world outcomes; and

training the second engagement machine learning model to output second engagement scores that minimize the margins of error.

11. A system for determining an engagement score of a user for a real-world entity represented by a virtual object in an extended-reality environment, comprising:

a processor; and

a computer readable storage medium having computer readable program code embodied therein that when executed by the processor performs operations, the operations comprising:

receiving, from tracking devices, movement parameters from a user in a real-world while the user is interacting with a virtual object in the extended-reality environment;

processing the movement parameters to determine a first engagement score indicating user interest in the real-world entity represented by the virtual object;

determining information on user information access requests with respect to the real-world entity;

inputting the information on the user information access requests to an engagement machine learning model to output a second engagement score indicating user interest in the real-world entity; and

outputting the first engagement score and the second engagement score to provide information on an efficacy of the virtual object in promoting interest in the real-world entity.

12. The system of claim **11**, wherein the tracking devices include a camera to record the movement parameters of the user moving in the real-world and an extended-reality headset to capture user eye movements during interaction with the virtual object in the extended-reality environment, wherein the determining the first engagement score comprises:

inputting the movement parameters and the user eye movement to a first engagement machine learning model to output the first engagement score, wherein the engagement machine learning model to output the second engagement score comprises a second engagement machine learning model.

13. The system of claim **12**, wherein the determining the first engagement score comprises:

calculating angular displacements, during engagement time, as a function of the movement parameters comprising handset controller movement coordinates and headset angles, wherein the movement parameters inputted to the first engagement machine learning model comprise the angular displacements.

14. The system of claim **11**, wherein the determining the information on the user information access requests with respect to the real-world entity comprises:

determining user questions entered into a virtual agent rendered in the extended-reality environment concerning the real-world entity; and

processing, by a large language model, the user questions to output summaries of key points of the user questions and user intent with respect to the real-world entity, wherein the summaries of the key points and the user intent are entered into the engagement machine learning model to determine the second engagement score.

15. The system of claim **11**, wherein the processing the movement parameters comprises inputting the movement parameters to a first engagement machine learning model to determine the first engagement score, wherein the engagement machine learning model to calculate the second engagement score comprises a second engagement machine learning model, wherein the operations further comprise:

receiving real-world outcomes with respect to users, for which the first and second engagement scores are generated, interacting with the real-world entity in the real world;

generating a first training set including movement parameters for multiple users interacting with the virtual object, first engagement scores for the multiple users, a quantized real-world outcomes, and margins of error between the first engagement scores and the quantized real-world outcomes; and

training the first engagement machine learning model to output first engagement scores that minimize the margins of error.

16. A computer implemented method for determining an engagement score of a user for a real-world entity represented by a virtual object in an extended-reality environment, comprising:

receiving, from tracking devices, movement parameters from a user in a real-world while the user is interacting with a virtual object in the extended-reality environment;

processing the movement parameters to determine a first engagement score indicating user interest in the real-world entity represented by the virtual object;

determining information on user information access requests with respect to the real-world entity;

inputting the information on the user information access requests to an engagement machine learning model to output a second engagement score indicating user interest in the real-world entity; and

outputting the first engagement score and the second engagement score to provide information on an efficacy of the virtual object in promoting interest in the real-world entity.

17. The method of claim **16**, wherein the tracking devices include a camera to record the movement parameters of the user moving in the real-world and an extended-reality headset to capture user eye movements during interaction with the virtual object in the extended-reality environment, wherein the determining the first engagement score comprises:

inputting the movement parameters and the user eye movement to a first engagement machine learning model to output the first engagement score, wherein the engagement machine learning model to output the second engagement score comprises a second engagement machine learning model.

18. The method of claim **17**, wherein the determining the first engagement score comprises:

calculating angular displacements, during engagement time, as a function of the movement parameters comprising handset controller movement coordinates and headset angles, wherein the movement parameters inputted to the first engagement machine learning model comprise the angular displacements.

19. The method of claim **16**, wherein the determining the information on the user information access requests with respect to the real-world entity comprises:

determining user questions entered into a virtual agent rendered in the extended-reality environment concerning the real-world entity; and

processing, by a large language model, the user questions to output summaries of key points of the user questions and user intent with respect to the real-world entity, wherein the summaries of the key points and the user intent are entered into the engagement machine learning model to determine the second engagement score.

20. The method of claim **16**, wherein the processing the movement parameters comprises inputting the movement parameters to a first engagement machine learning model to determine the first engagement score, wherein the engagement machine learning model to calculate the second engagement score comprises a second engagement machine learning model, wherein the operations further comprise:

receiving real-world outcomes with respect to users, for which the first and second engagement scores are generated, interacting with the real-world entity in the real world;

generating a first training set including movement parameters for multiple users interacting with the virtual object, first engagement scores for the multiple users, a

quantized real-world outcomes, and margins of error between the first engagement scores and the quantized real-world outcomes; and
training the first engagement machine learning model to output first engagement scores that minimize the margins of error.

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